MIT 8103: ADVANCED DATABASE SYSTEMS

Practical Coursework – Final Portfolio Documentation

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Course: MIT 8103 - Advanced Database Systems

Project: Luma Polytechnic Database Portfolio

1. Introduction and Case Study

This project utilizes all the knowledge we've learned about the advanced topic of database systems to model a Nigerian Polytechnic called Luma Polytechnic. Luma Polytechnic is a simulated Nigerian Polytechnic. The details covered include: database modelling and design for the academic's section of the polytechnic which includes students, departments, programs, lecturers, courses, semesters, course offerings, enrolments and results. The relational database was set up using MySQL compatible SQL on XAMPP with MariaDB/phpMyAdmin for the relational part and MongoDB Atlas with MongoDB Compass for the Nosql/Cloud based assignments. All the source codes, analysis, tests scripts and evidence checklists were on GitHub.

2. The Five Portfolios

Portfolio 1: Database Design and Modelling

This portfolio shows that there are 9 core tables that are related through the use of primary and foreign keys.

Constraints are examined using foreign key constraints, indexes and testing for invalid data. Check constraints were used to restrict CAS scores between 0 and 30 and Exam scores between 0 and 70.

There were examples of how grade records show both Total Score and Letter Grades and what Grade Points they scored. This portfolio illustrated how constraints enforce integrity throughout the database.

Portfolio 2: Query Processing and Optimisation

This portfolio uses Explain, Joins, Filters and Indexes to illustrate the processing of queries.

Student information query used the Unique index on matric no and Grade information query used the index on Total Score to obtain the desired results. This demonstrates how the indexes assist the DBMS in the Query Execution.

However as our database is relatively small and execution times were very short we were unable to establish any real improvements in performance and hence concentrated on how the Explain would return the index usage rather than concentrating on the actual time taken.

Portfolio 3: Transactions and Concurrency

This portfolio examines Transaction control and Isolation in the MariaDB shell.

Rollback of an uncommitted Balance transaction was tested. Commit of a Balance transaction was also tested. In a Balance transfer transaction, both 'from' and 'to' accounts were changed from 1000.00/500.00 to 900.00/600.00 and the resultant committed values were noted.

In a Repeatable Read scenario where two transactions are run concurrently, one transaction sees its own uncommitted Change while the other transaction continues to see the previous Committed values until it itself is Committed. The reported Isolation level was REPEATABLE READ and attempting to set a Blocking Select. For Update transaction failed to achieve the desired result.

Portfolio 4: NoSQL and Advanced Data Models

A "students" MongoDB collection was created in the luma_nosql database on MongoDB Atlas. Embedded details of each student's registration and grades were stored in each document. Finding students who earned an "A" grade and an aggregation to sum all registration scores were accomplished with queries.

For the four documents shown in the aggregation output, the mean total score was 72, the high was 87, and the low was 53. An ascending index was added to matric_number. The explain showed an index scan followed by document retrieval.

However, because there are only four documents in this collection, no conclusion could be drawn about the performance benefits of indexing.

Portfolio 5: Distributed and Cloud Database Exercise

MongoDB Compass was able to connect to the luma_nosql database hosted on MongoDB Atlas and retrieve the student record with the matric number LP/2025/001.

The MongoDB command "hello" reported which replica set the Compass client was talking to and told us that we were currently talking to the writable primary node. Replica set status, as shown in the provided excerpt, revealed a primary and a secondary member, both with a health of 1. This shows that we have connectivity to our cloud provider and also a basic understanding of how replication works. We did not attempt a failover, so we don't count this as an experimentally verified capability.

3. Technical Reflection.

As I already knew SQL and Relational databases, I was able to work with relational database systems using the knowledge I already had. For example, I was able to understand and do relational database designs and optimizations using the knowledge I already had. But I was also new to NoSQL databases and had a lot to learn. In particular, I had no experience with

MongoDB at all. So I have to learn how it works storing data as "document" inside collections (and not storing everything in tables as relational databases). And I was able to use MongoDB Compass and MongoDB Atlas to do queries on documents, to embed registration and grades details inside student documents and to use filters and aggregation pipelines to perform some analysis.

I also created an index over matric_number and studied the Explain results to see that indexes are still important even for document oriented databases, though the way they are applied is different. Also, MongoDB Atlas brought my attention to the fact that cloud hosted databases are a reality now and performing a ReplicaSet inspection helped me to understand better Primary and Secondary roles of them and how they help with high availability. As we had a really small dataset, I could not compare very well performance and do some other things I was thinking about doing.

In the future, I would try to use a lot of more records and to simulate a controlled failover to understand better replication in more detail.

4. Ai-Use Declaration

AI role in this project was solely in assisting in structuring and organizing the portfolio documents, the DB design, SQL and MongoDB implementation, testing and determination of results was carried out by me and I am accountable for them.

5. Repository and Supporting Evidence

GitHub repository: https://github.com/denireal/advanced-database-portfolio.git This repository contains 5 portfolios of database design, query optimization, transaction & concurrency, NoSQL and Distributed/Cloud DBMS. Screenshots related to supporting evidence must be added inside respective folders before submitting.